UNIVERSITÀ DEGLI STUDI DI MILANO

FACOLTÀ DI SCIENZE E TECNOLOGIE

Design and Implementation of an Autonomous Robotic System for 3D Scanning

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"Every milestone achieved is merely proof that I could do more"

Abstract

This thesis is situated within the field of precision agriculture, developed in collaboration with the AIS Lab of the University of Milan and the Department of Agricultural Sciences. The primary objective of the research is the design and integration of an advanced robotic infrastructure for crop sensing, based on the use of a six-axis cobot from the Dobot line, equipped with a 3D stereoscopic camera installed on the end-effector.

The original contribution of this thesis lies in the integration of the MoveIt framework for managing the kinematics and motion planning of the manipulator, aimed at implementing a dynamic scanning pipeline. This system enables the robot to acquire morphological data of plants from multiple viewpoints, overcoming the limitations of static vision systems. The developed software architecture ensures bidirectional communication between the robot control system and the vision sensor, allowing for the generation of optimized trajectories based on perception of the surrounding environment.

Experimental tests conducted on the prototype confirmed the effectiveness of motion planning in ensuring comprehensive coverage of the scanned surface. Despite some inherent limitations in the mechanical structure of the cobot, the system demonstrated remarkable automation capability in the data acquisition process. In conclusion, this work demonstrates how the synergy between collaborative robotics and 3D vision can elevate monitoring standards in protected crop environments, laying the groundwork for future developments in adaptive path planning optimization and integration of SLAM techniques for real-time pose refinement.

Contents

Chapter 1

Introduction

1.1 Automated Crop Sensing

Importance of Automation in Modern Agriculture

In recent years, the agricultural sector has undergone a profound technological transformation, marked by the introduction of advanced solutions in processes historically characterized by high manual labor requirements. Among these innovations, automated crop sensing plays a role of paramount importance. The integration of robotic systems enables evolution from management based on statistical averages to a "plant-by-plant" approach, allowing for reduced chemical treatments and optimized water resource usage. In this transition toward Agriculture 4.0, the robot is no longer viewed as a mere operating machine, but as an intelligent and interconnected node, capable of acquiring and processing data in real-time to support complex agronomic decisions.

Impact of Robotic Sensing Technologies on Production Efficiency

Robotic sensing technologies represent the fundamental pillar for phenotypic monitoring and analysis of crop status. The use of advanced sensors for high-throughput phenotypic monitoring[1], such as stereoscopic cameras and depth sensors installed on robotic arms, enables overcoming the intrinsic limitations of human vision and static monitoring systems. Production efficiency is enhanced through the acquisition of high-precision morphological data, essential for early identification of water stress, nutritional deficiencies, or pathogenic events. Automation of this process ensures data repeatability and sampling frequency otherwise unattainable, translating visual observation into quantitative parameters ready for analysis.

Technological Challenges in Implementation of Autonomous Systems

Despite its high potential, the implementation of autonomous systems in agriculture presents significant technical challenges, arising primarily from the unstructured nature of the operational environment. Unlike conventional industrial contexts, where objects are rigid and standardized, agricultural robotics must contend with extreme morphological variability of plants. From the computational and control perspective, the main criticalities lie in maintaining the optimal distance between sensor and target necessary to ensure data quality, and in the safety of motion planning. The latter is crucial to avoid accidental collisions that could damage both the instrumentation and the observed plant organism.

1.2 Problem Definition

The project addresses the need of the Department of Agricultural Sciences at the University of Milan to monitor the growth of small plants in a controlled environment without the need for manual acquisitions, which could result in imprecision. To this end, a Dobot CR5 cobot equipped with a ZED 2i stereoscopic camera was used.

1.3 Thesis Objectives

The primary objective of the thesis is the development of an integrated architecture composed of three fundamental interconnected modules. First, the integration of the proprietary software system of the robot with the ROS infrastructure and movement-focused frameworks was realized. In parallel, the work involved creating a communication module to interface the 3D camera with the Dobot CR5 cobot. Finally, an algorithm was designed to generate optimized trajectories based on three-dimensional perception of the environment. Such a structured system enables addressing criticalities linked to the manipulator's kinematics and the morphological variability of plants, which change in position and dimensions over time. Integration of these modules allows the robot to dynamically adapt its movement, ensuring precise and consistent data acquisition over time.

1.4 Thesis Structure

The thesis work is organized into seven main chapters, in addition to the introduction:

- **Chapter 2** analyzes the state of the art and enabling technologies. Examination includes the ROS2 and MoveIt2 ecosystems for manipulator management, current 3D vision techniques applied to agriculture, and methodologies for three-dimensional environment reconstruction and autonomous viewpoint planning.

- **Chapter 3** describes the specific technologies and methodologies used in the project. It deepens the characteristics of the ROS2 middleware, the MoveIt2 motion planning framework, and the multi-level scanning strategy adopted for optimizing visual coverage of plants.

- **Chapter 4** describes the design and implementation of the robotic control system. This section details the reverse engineering activity on the official Dobot CR5 drivers, establishes necessary requirements that the new system must meet, and reports structural modifications made to comply with them.

- **Chapter 5** presents the design of the integrated scanning system. The focus shifts to communication between the cobot and the ZED 2i sensor, describing data exchange architecture, information preprocessing logic, and trajectory planning for each individual plant.

- **Chapter 6** is dedicated to testing and validation activities. Results obtained in both simulated and real-world system environments are reported, comparing possible discrepancies, critically analyzing the precision of scanning point achievement, and structural limitations encountered during operational testing.

- **Chapter 7** gathers the conclusions of the work. A synthesis of results achieved relative to the set objectives is offered, and future development perspectives are outlined, with particular reference to integration with mobile robotic bases and improvement of system adaptability to complex agricultural scenarios.

Chapter 2

State of the Art and Technologies

This chapter analyzes the technological and methodological foundations on which this thesis work is based. The objective is to provide a comprehensive overview of existing solutions for robotic manipulator control, three-dimensional perception, and autonomous movement planning, with specific focus on their applications in the precision agriculture sector. The ROS2 ecosystem and MoveIt2 framework will be examined, followed by an in-depth exploration of 3D vision techniques and autonomous exploration strategies that enable advanced crop sensing systems.

2.1 ROS2 Ecosystem and MoveIt2 for Manipulators

Modern robotics, especially collaborative and service robotics, relies on standardized software platforms to reduce development complexity and promote interoperability. In this context, ROS2 (Robot Operating System 2) has established itself as the *de facto* standard.

2.1.1 ROS2 Architecture: DDS, Nodes, and Lifecycle

ROS2 represents a profound redesign of its predecessor[10], introducing the **Data Distribution Service (DDS)**[12] as the default communication middleware, based on the RTPS protocol[13]. This industrial standard ensures real-time, reliable, and scalable communication, overcome the limitations of the centralized manager (*roscore*) of ROS1. The architecture is based on a distributed system of independent processes, the **nodes**, which communicate through *publish/subscribe*, *service/client*, and *action* mechanisms. A crucial innovation is the introduction of a **managed node lifecycle** (Managed Nodes), which allows control of the state of software components (configuration, activation, deactivation), increasing robustness and reliability of complex systems.

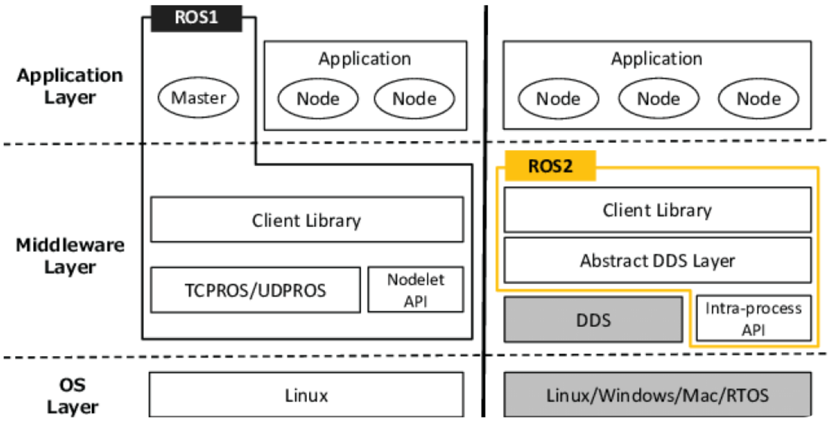

Figure 2.1: Architectural comparison between ROS1 and ROS2.

2.1.2 MoveIt2: Planning, Collision Detection, and Planning Scene

For robotic manipulator control, **MoveIt2**[18] is the most widespread motion planning framework in the ROS2 ecosystem, based on the OMPL library[19] for planning algorithms. It provides a unified interface for planning complex trajectories, inverse kinematics (IK), control, and collision prevention. The heart of MoveIt2 is the **Planning Scene**, a virtual representation of the robot's work environment. It includes the kinematic model of the robot (URDF), the real-time state of its joints, and a map of obstacles, which can be defined statically or detected dynamically by sensors. Thanks to this representation, planning algorithms, such as RRT-Connect[9] based on Probabilistic Roadmaps[6], can generate safe trajectories that avoid collisions using FCL for collision checking[14] with both the external environment and the robot itself (*self-collision*), a requirement necessary for safe manipulation near delicate plants.

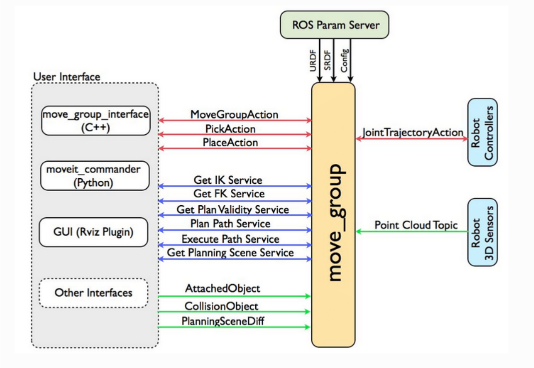

Figure 2.2: Overview of MoveIt2: main components and planning pipeline.

2.1.3 Sensor Integration and Control Interfaces

2.2 3D Vision Techniques for Agriculture

Three-dimensional morphological data acquisition is a cornerstone of precision agriculture. As highlighted by various studies in the field, including reviews of 3D vision techniques for agricultural applications, the choice of sensing technology depends on the trade-off between accuracy, cost, and operating conditions.

2.2.1 Multiview Systems: Stereo and Structure-from-Motion

Stereoscopic cameras, such as the ZED 2i used in this project, represent a versatile and effective solution. They mimic human binocular vision, calculating depth through **triangulation** of corresponding points between two images acquired from known viewpoints[16]. This technology offers a good balance between point cloud density and acquisition speed. A related technique is **Structure-from-Motion (SfM)**[4], which reconstructs scene geometry from a series of 2D images acquired from different positions, simultaneously estimating the 3D structure and camera parameters. Although powerful, SfM is computationally intensive and more suitable for offline reconstruction.

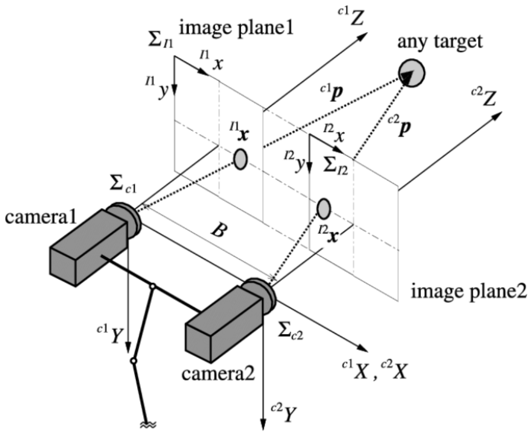

Figure 2.3: Stereo vision principle: triangulation for depth estimation.

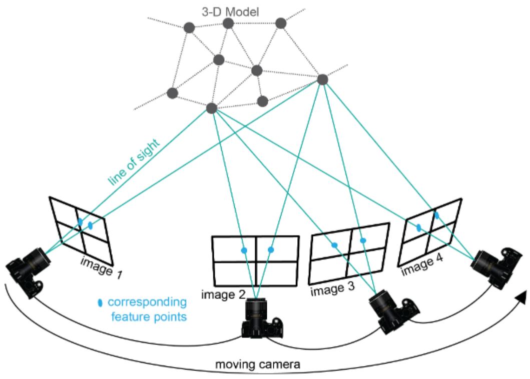

Figure 2.4: Simplified Structure-from-Motion (SfM) pipeline.

2.2.2 Time-of-Flight (ToF) and LiDAR Sensors

Time-of-Flight (ToF) and **LiDAR (Light Detection and Ranging)** sensors are *active sensing* technologies. LiDAR measures distance by calculating the time taken by a laser pulse to reach an object and return to the sensor. This technique is extremely precise and robust to lighting conditions, but often more costly. ToF operates similarly, but using a modulated light source. Both technologies are widely used in agricultural robotics for terrain mapping and biomass estimation.

2.3 3D Environmental Reconstruction and Autonomous Navigation

To operate autonomously, a robot must be able to build a map of the environment and localize itself within it. This problem is addressed by **SLAM (Simultaneous Localization and Mapping)** techniques.

2.3.1 SLAM: Overview and Variants

SLAM is a fundamental concept for autonomous navigation, as explored in construction and logistics contexts based on LiDAR. SLAM algorithms, whether based on filters (e.g., EKF) or graph optimization, enable a mobile robot to operate in unknown environments. In agriculture, visual SLAM (vSLAM) or LiDAR-based SLAM is crucial for navigation between crop rows and crop mapping, enabling an AGV to position a manipulator like the Dobot CR5 with precision.

2.3.2 Integration with NeRF and Gaussian Splatting

3D reconstruction techniques are evolving rapidly. **Neural Radiance Fields (NeRF)**[11] and **3D Gaussian Splatting**[7] represent the state of the art for *novel view synthesis*, allowing for the derivation of photorealistic three-dimensional representations from sets of RGB images. Although not implemented in this work, these methodologies constitute a natural future extension for high-precision phenotyping applications, for example for accurate leaf volume estimation or fruit counting.

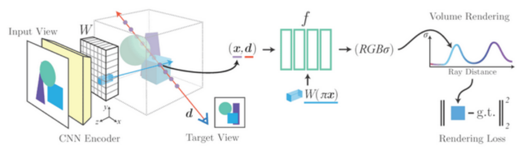

Figure 2.5: Convolutional NeRF architecture.

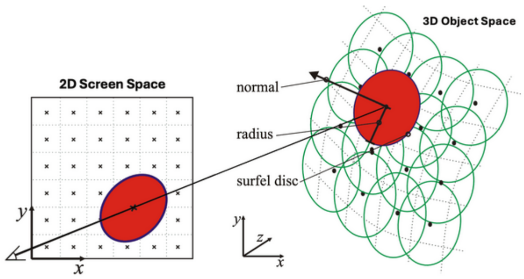

Figure 2.6: 3D representation with Gaussian Splatting.

2.4 Autonomous Viewpoint Planning

Once the robot is localized and has a partial map of the environment, the problem arises of deciding where to look to maximize information acquisition. This is the field of autonomous viewpoint planning, or **Next-Best-View (NBV)**.

2.4.1 Problem Modeling and Algorithm Classes

The NBV problem consists of selecting the next sensor pose (in this case, the camera on the end-effector) that maximizes an objective function[8], such as uncertainty reduction or increased explored surface area. As described in the literature on exploration strategies, classical approaches are based on space discretization (e.g., *voxel grid*) and geometric heuristics for "looking" toward unexplored areas. More recent methodologies, instead, exploit reinforcement learning (*Deep Reinforcement Learning*) to train agents capable of developing complex and adaptive exploration strategies, surpassing the performance of rule-based approaches, especially in unstructured environments like agricultural ones. Integration of such algorithms represents one of the most promising future directions for this work.

Although these approaches offer great adaptability, they require substantial computational resources and extensive datasets. In application contexts where target geometry is known (such as the vertical structure of plants), deterministic geometric approaches offer an optimal compromise between reliability and execution speed.

Chapter 3

Technologies and Methodologies Used

This chapter examines the technological choices and methodologies adopted for developing the control and scanning infrastructure. The first part analyzes the software architecture implemented for controlled movement of the Dobot CR5 cobot, highlighting the design decisions that guided its integration. The second section explores the algorithmic logic developed for generating optimal scanning paths.

Figure 3.1: Dobot CR5 used as manipulation platform.

3.1 Control Infrastructure and Motion Planning

This section describes in detail how the ROS2 ecosystem and MoveIt2 framework were configured and adapted to meet the specific requirements of the project.

Architectural Choices and Hardware Integration

The choice to adopt **ROS2 humble** as middleware was driven by its distributed DDS-based architecture, which proved fundamental for ensuring low-latency and high-reliability communication among the control system, the manipulator, and the 3D sensor. This modularity allowed decoupling of perception logic from actuation logic, a key requirement for a maintainable and scalable system.

Integration of the **MoveIt2** framework was the response to the need to overcome the limitations of native Dobot CR5 drivers, which offer basic positional control but lack advanced planning and collision management capabilities. The adopted methodology therefore focused on creating a software "bridge" linking the high-level planning of MoveIt2 with the robot's low-level controller, transforming the arm from a simple actuator into an intelligent tool aware of its own environment.

3.2 Scanning Methodology for Visual Coverage Optimization

Once the movement infrastructure was established, a dynamic scanning methodology was defined to maximize data quality acquired through the ZED 2i camera. The methodological approach is structured in three sequential logical phases:

Perception and Prioritization

The algorithm receives data acquired by the stereoscopic camera and reprocesses the environment to identify present plants. In this phase, the system establishes a scanning priority order, determining which subject to analyze first based on spatial arrangement.

Multi-Level Planning

For each identified plant, the system estimates its volumetric dimensions. Based on these data, the algorithm calculates the necessary number of circular scans (orbits) to be performed at different heights, ensuring that every section of the plant, from base to top, is adequately covered.

Figure 3.2: Multi-level planning: scanning orbits at different heights.

Optimal Viewpoint Selection

For each orbit, the algorithm determines a series of observation points on the circumference. Being an adaptive system, point selection is not rigid: the software evaluates the surrounding environment and autonomously chooses optimal positions offering the best target visibility, discarding points that are kinematically unreachable or unnecessary because already covered by other acquisitions.

Chapter 4

Design and Implementation of the Robotic Control System

This chapter describes the robot movement system design process, structured in four main phases. Given the project's nature, focused on ROS2 integration in a pre-existing ecosystem, the treatment will initially analyze the base system and its limitations, then define requirements for the proposed extension, the architecture of developed modules, and the automated activation pipeline.

4.1 Base System Analysis

The reference hardware system is the Dobot CR5 cobot. The manufacturer natively provides a suite of scripts for activating drivers and low-level APIs for arm interaction, such as movement toward Cartesian coordinates or speed management. These APIs, however, operate with limited logical control: if a point is kinematically complex or unreachable (singularity or outside workspace), the system fails execution without providing recovery strategy. An infrastructure capable of calculating optimal and safe trajectories is therefore necessary. Although the manufacturer provides a repository for ROS2 interfacing, it is limited to simulation only; only components necessary for communication between the Dobot mock on Rviz[1] and MoveIt2 are provided, such as:

- URDF/XACRO[2]: The repository provided the geometric and kinematic

[1]RViz (ROS Visualization) is the fundamental 3D graphical interface in the ROS ecosystem. It allows real-time visualization of the robot's virtual model (URDF), data from sensors (such as point clouds from the ZED camera), trajectories planned by MoveIt2, and the entire Planning Scene, facilitating system monitoring and debugging.

[2]The Unified Robot Description Format (URDF)[20] is an XML file used to describe robot physical elements. XACRO is a macro language that allows such descriptions to be modular

models necessary for MoveIt2 to "know" the robot structure.

- Launch Files[3]: Scripts were present for launching only the simulated environment.
- Various scripts for MoveIt2 activation

. For physical robot operation, a Hardware Interface generally must be integrated[4]: This component, fundamental for translating the movement plan into electrical signals for real motors, was absent in the original version. The thesis work thus filled the technological gap by creating a software "bridge" between the planning engine and the Dobot's physical controller.

4.2 Extended System Requirements

For the system to be considered complete and efficient, the proposed architecture must satisfy the following requirements:

- Reachability and Robustness: Ensure achievement of any physically accessible point in the workspace within acceptable times, eliminating intermediate failures and preventing "crash" states of the native controller.
- Obstacle Management: The system must integrate environmental perception into planning, calculating paths that avoid collisions with objects present in the scene.
- Startup Automation: To reduce human error and simplify user experience, the system must provide an automated startup procedure that manages sequential activation of all modules.

4.3 Proposed System Architecture

To fully comply with previously imposed requirements, the developed architecture is based on a communication module serving as a "translator" between FollowJointTrajectory messages sent by MoveIt2 and Dobot proprietary APIs. MoveIt2 decomposes a complex trajectory into a series of intermediate points (waypoints) easily reachable.

and readable.

[3]Launch files in ROS allow simultaneous startup of multiple nodes and system parameter configuration with a single command.

[4]The Hardware Interface is the software component serving as an abstraction layer between ROS software commands (e.g., desired velocities or positions) and physical drivers controlling robot motors.

The developed module forwards these points sequentially to the robot controller. To ensure smooth movement and avoid jerking between points, the Continuous Path function was implemented[5]. Subsequently, a launch file comprehensive of all components necessary for actual operation was created:

- **Static TF and TF:** Manage fixed and mobile transformations between reference frames (e.g., robot base, camera, end-effector).
- **Move Group:** The central MoveIt2 node coordinating planning capabilities.
- **Translator Node:** The previously described translation component for controller communication.
- **ZED Components:** Components for proper perception-execution communication addressed in subsequent chapters.

Finally, safety is ensured by use of MoveIt2's Planning Scene[6]. Through native functions, detected objects are included in the robot's virtual world, making trajectories kinematically safe relative to obstacles.

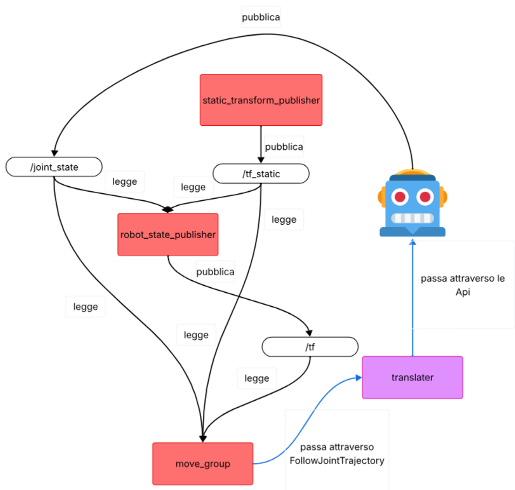

Figure 4.1: Architecture of translation module between MoveIt2 and Dobot controller.

[5]Continuous Path allows the robot to move through a series of waypoints without complete stops at each one, coordinating trajectories to maintain constant, smooth velocity.

[6]The Planning Scene is the internal representation of the surrounding world maintained by MoveIt2. It integrates the kinematic model of the robot with sensor data and geometric objects (collision objects) inserted in the workspace. This representation is fundamental for collision detection algorithms, as it allows the planner to verify absence of intersections between the volume occupied by the robot along a trajectory and obstacles in the environment.

4.4 Operating Pipeline

Due to system complexity, a Bash script for orchestrated startup management had to be developed. The procedure follows a rigorous temporal sequence:

1. Network Configuration: The script automatically sets the operating machine's IP address to ensure correct communication with the Dobot controller on the same subnet.

2. Driver Activation: The native manufacturer driver script is started, necessary for proper Dobot configuration.

3. ROS2 Infrastructure Startup: Only after confirming correct driver startup is the launch file described in the previous section executed.

This centralized approach ensures that communication among camera, planner, and robot is synchronized, minimizing risks of critical error during initialization.

Chapter 5

Design and Implementation of the Scanning System

This chapter analyzes scanning system implementation, with particular attention to communication methodologies between the 3D sensor and the planning module. The data structures designed for environmental representation and the algorithm developed for generating optimal scanning trajectories are described.

5.1 Scanning System Requirements

To ensure high-quality data acquisition and smooth operation, the following functional requirements were established:

- **Real-time Communication Module:** Implementation of data exchange infrastructure between the ZED 2i camera and ROS2 nodes to ensure low-latency information flow.
- **Environmental Mapping:** Generation of spatial representation based on depth information, converted into a format suitable for movement planning.
- **Adaptive Planning:** System capability to establish the best trajectory for each individual plant, maximizing spatial coverage based on geometry detected by the sensor.

5.2 Proposed System Overview

The scanning system architecture is structured in three phases. Initially, the communication protocol between the ZED 2i camera and robot was defined through formalization of custom ROS messages. Subsequently, the data transmission channel and script for data reprocessing was realized. Finally, a coordination script was developed that, processing environmental data, calculates and executes optimal trajectories for the robotic arm.

5.3 Data Communication System Architecture

System design began with analysis of Dobot CR5 kinematics. Given one-meter maximum extension and six degrees of freedom of the ground-anchored arm, complete coverage of a plant organism proves complex. An approach based on circular trajectories with variable radius around the target was therefore chosen. For larger plants, the system automatically increases the number of orbits at different height levels. Current implementation follows a deterministic-dynamic approach: the system performs initial environmental acquisition to identify plant number and position, adapting the scanning routine to each startup. It is, however, assumed that the environment remains static during individual scanning session execution.

5.3.1 Message Definition and Preprocessing

Communication between sensor and robot occurs through ROS *Topics*[1]. Since the sensor provides raw scene data, a message hierarchy was designed to structure information:

- `Point.msg`: Fundamental Cartesian coordinates (x,y,z).
- `BoundingBox.msg`: Defines object spatial extent through *low_left* and *top_right* vertices.
- `OptimalPoint.msg`: Represents a scanning point on the circumference, including an *optimality* value to indicate reachability.
- `Circumference.msg`: An array of `OptimalPoint` composing an entire scanning orbit.
- `Object.msg`: Contains plant data (target or obstacle), its extent, and calculated circumferences.

[1] A Topic in ROS is a bus-based communication channel based on the publish/subscribe model. A node sends a message by publishing it to a specific topic, while interested nodes receive data by subscribing to it.

- `Map.msg`: The root message including workspace limits and list of all detected objects.

Data processing is handled by the `reworked_map.cpp` script. Given a variable radius, it calculates necessary orbits to cover the plant in height and generates circumference sampling points. For each point, the script verifies compatibility with robot workspace and absence of collisions with other objects: if a point is physically unreachable or inside an obstacle, the *optimality* parameter is set to zero; all processing is executed on each plant.

5.3.2 3D Sensor Communication

To interface vision software with the robot ecosystem, **rosbridge** was used[2][21]. This allowed the camera to act as *publisher*, sending environmental coordinates already transformed into the Dobot spatial reference frame.

5.4 Trajectory Planning System Architecture

Upon receiving structured data, the system interacts with the planning module according to obstacle addition and trajectory processing phases.

5.4.1 Adding Obstacles to the Planning Scene

To ensure movement safety, each detected object (plants or fixed structures) is inserted into MoveIt2's *Planning Scene* as a prismatic solid (*box*). This allows the algorithm to avoid collisions during passage between various scanning points.

5.4.2 Trajectory Processing

The system iteratively processes each plant. For each circumference, **boundary points** are identified, namely reachable points adjacent to inaccessible zones. The arm attempts to reach the nearest boundary point; should inverse kinematics fail, the point is discarded and proceeds to the next optimal point. Once positioned, the robot orients the camera toward the plant center and performs acquisition. Upon completion of each shot, points within the camera's field of view (FOV) are marked as scanned to ensure complete but non-redundant scans, and the process continues until entire orbit coverage and, subsequently, all plant levels.

[2]Rosbridge provides JSON interface for ROS commands, allowing external programs or systems not natively running ROS to publish and subscribe to topic messages.

5.5 Operating Pipeline

The scanning system operational sequence is structured as follows:

1. The robot controller initializes the *rosbridge* channel.
2. The 3D sensor connects to the bridge, publishes environment raw coordinates on the `/boing` topic.
3. The preprocessing node extracts data from `/boing`, performs spatial reprocessing and orbit calculation, publishing the final map on the `/reworked_map` topic.
4. The coordination script receives the map and initiates the trajectory execution pipeline via MoveIt2.

Chapter 6

System Testing and Validation Activities

This chapter describes the validation process of the developed architecture, aiming to verify scanning pipeline effectiveness and hardware-software integration robustness. The testing procedure was structured in two sequential phases: initially the system was validated in a controlled virtual environment, then progressing to experimentation on the physical robot. The adopted methodology is based on monitoring software behavior in various spatial configurations, analyzing system response capability in both favorable scenarios and limiting conditions.

6.1 Testing in Simulated Environment

Preliminary validation in simulated environment was fundamental for isolating the scanning algorithm from physical variables and latencies of the real communication module. For this phase, the `RViz` visualizer was used[1], which enabled real-time analysis of trajectory generation by MoveIt2.

[1]RViz (ROS Visualization) is ROS's primary tool for 3D visualization of robot state, sensor data, and planned trajectories, allowing monitoring of the Planning Scene and motion planning software behavior.

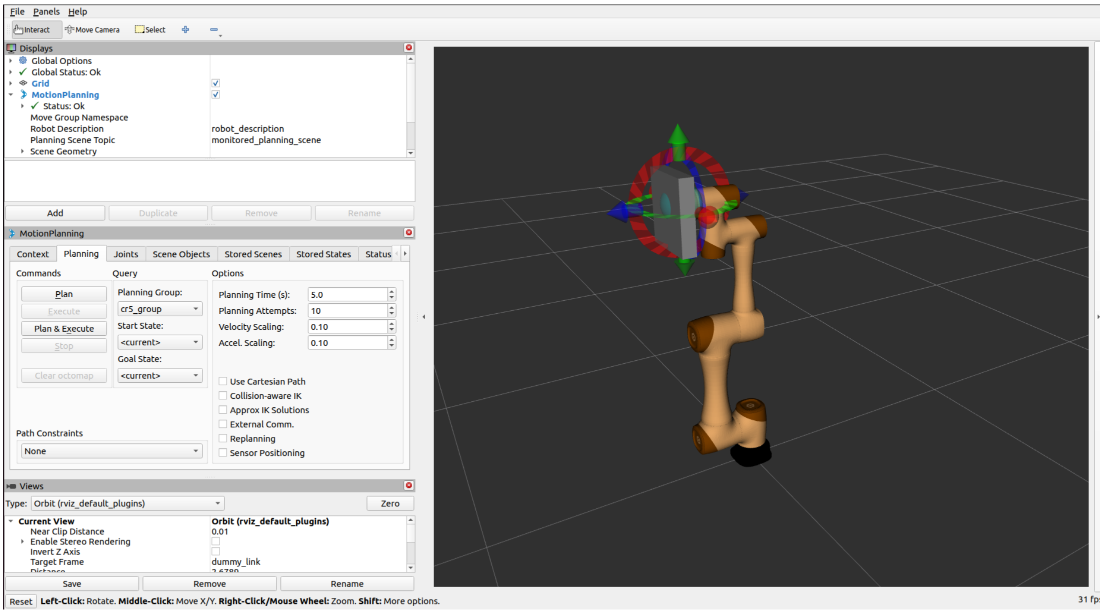

Figure 6.1: Test configuration in simulated environment with Dobot CR5.

6.1.1 Validation Objectives in Simulation

Performance evaluation in simulation was based on two primary metrics:

- **Spatial Coverage:** Verification of scanned surface percentage for each plant in relation to its workspace position.

  In the scanning module code (`PlantScanner`), this metric is defined as the percentage ratio between actually acquired points and the complete set of spatial points generated for the plant. In particular, the implemented formula is:

$$\text{Coverage\_\%} = \begin{cases} 100 \times \dfrac{N_{\text{covered}}}{N_{\text{total}}} & \text{if } N_{\text{total}} > 0, \\ 0 & \text{otherwise.} \end{cases}$$

  Where terms are defined as follows:

  - N_{total}: the total number of elements in the scanning points vector (`scan_points`) calculated on plant circumferences. This set is all-inclusive and includes even points with optimality $= 0$ parameter.
  - N_{covered}: the number of points whose `covered` flag is true upon completion of the entire scanning procedure.

Implementation Note: The algorithm allows marking as `covered` even points with optimality = 0 if located within distance radius less than `CAMERA_FOV_HORIZONTAL` from the scanning pose, based on a spherical FOV model. Consequently, the metric reflects the fraction of points actually acquired relative to the entire set of points planned for the plant, not just initially considered reachable.

- **Computational Efficiency:** Analysis of number of acquisitions performed and validity of observation points selected by the algorithm.

Testing with Multiple Plant Configurations

System capacity to handle complex scenes characterized by simultaneous presence of multiple target objects was tested. This phase allowed verification of algorithm prioritization logic and effectiveness of sequential scanning without accuracy loss between instances.

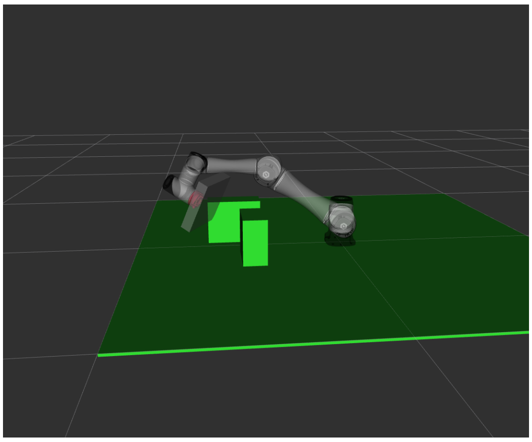

Figure 6.2: Scanning of multiple plants in a complex configuration.

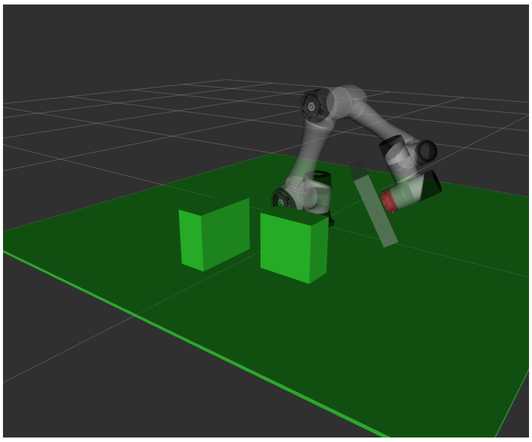

Figure 6.3: Scanning of multiple plants in a complex configuration.

6.1.2 Results and Critical Observations

Simulations conducted on variable configurations (from one to four objects, including obstacles) confirmed algorithm stability, reporting no critical failures or manipulator error states.

- **Scanning Performance:** For each plant positioned in optimal workspace area, the system averages between seven and eight acquisitions per orbit, ensuring coverage between 90% and 100%.

- **Timing and Planning:** Scanning duration ranges to a few minutes per plant. The planning algorithm is observed to privilege single trajectory calculation speed over global movement sequence optimization, a compromise acceptable given the application context.
- **Collision Avoidance:** All generated trajectories successfully avoided collisions, confirming Planning Scene integrity.

Analysis of Kinematic Limitations

Anchoring the Dobot CR5 base to the floor level inherently limits useful work volume, making scanning difficult in areas immediately adjacent to the base or at extreme workspace edges. A possible design solution to extend mobility would consist of robot installation in suspended configuration (anchored to upper support), permitting freer zenith and lateral vision. However, even with these limitations, the system demonstrated satisfactory accuracy (70-80%) in peripheral zones, still ensuring homogeneous plant reconstruction.

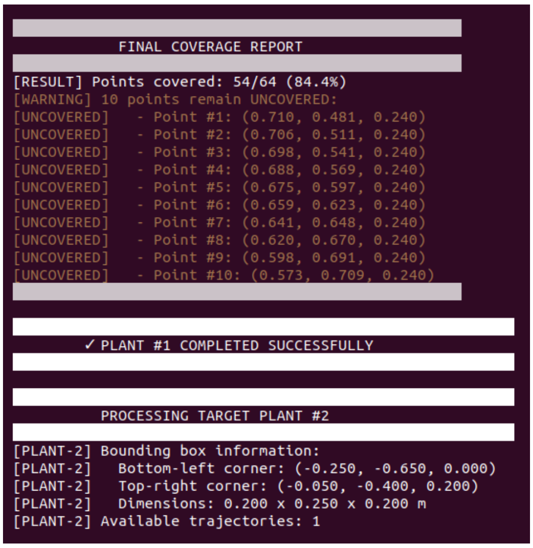

Figure 6.4: Coverage of plant positioned in partially optimal workspace area.

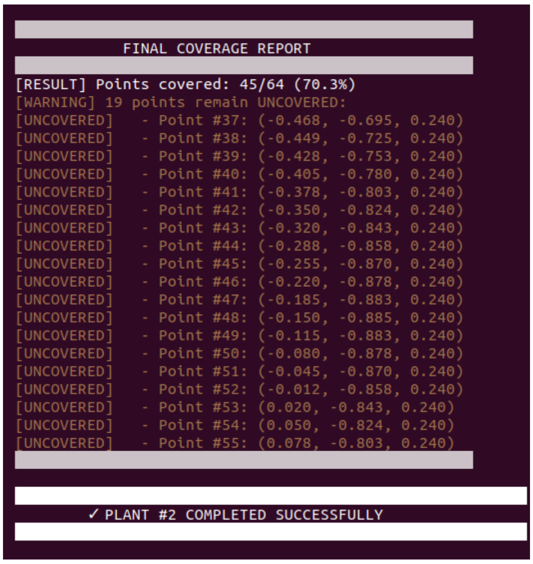

Figure 6.5: Coverage of plant positioned in non-optimal workspace area.

6.2 Testing with Real System

Upon completion of the virtual phase, the system was tested on physical hardware to detect possible discrepancies between the simulated model and operational conditions.

6.2.1 Validation Objectives in Real Environment

A test is considered passed if the real system faithfully replicates behavior observed in simulation, completing initialization pipeline without errors and ensuring absence of physical collisions with the surrounding environment.

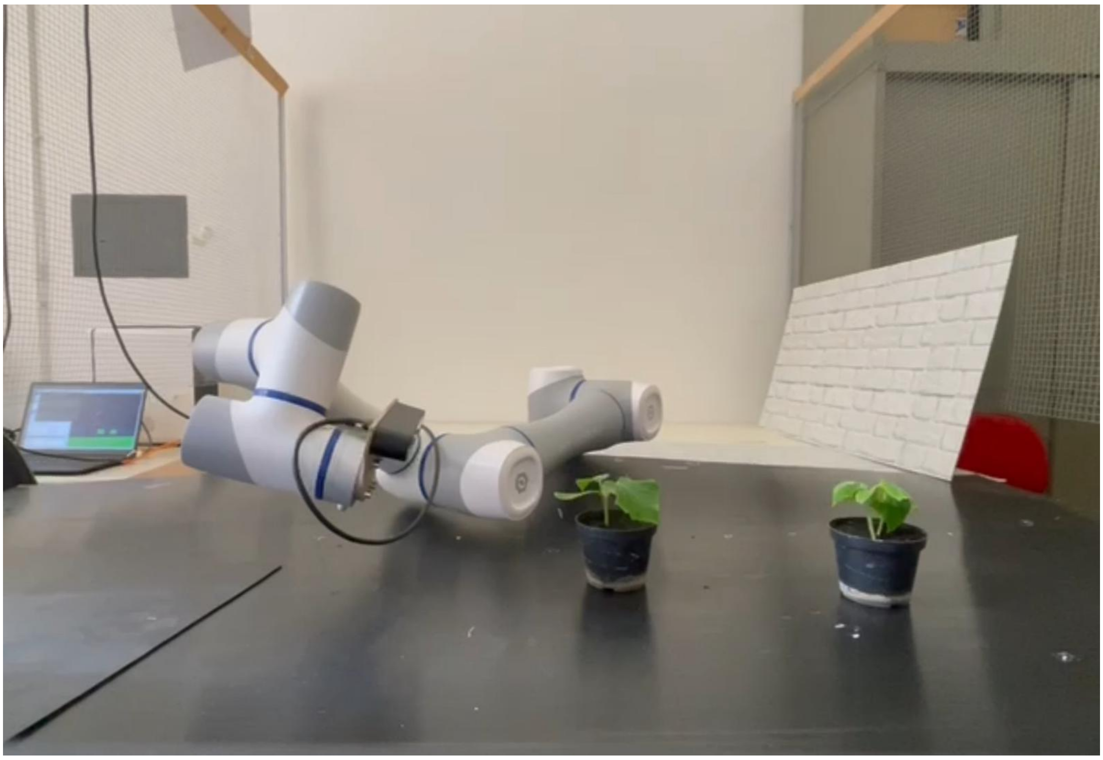

Figure 6.6: Test configuration in real environment with Dobot CR5.

Verification of Simulation-to-Reality Correspondence

Comparison between the two environments revealed optimal results and substantial coherence in scanning accuracy. However, minor discrepancies emerged:

- **End-Effector Precision:** Rarely, some scanning points valid in simulation were not executed on the real robot. This is attributable to minimal millimetric discrepancies in the CAD model of the camera module mounted on the robot extremity. Such deviation did not impact final coverage thanks to conservative choice of reduced Field of View (FOV), ensuring greater error tolerance.

- **Sensor Calibration:** Marginal errors in detected object dimensions were traced to limitations in intrinsic calibration of the ZED 2i camera vision module, independent of developed software logic.

6.2.2 Results and Performance Comparison

From the performance perspective, real robot movement speed resulted 20% lower than simulation; this limitation was deliberately imposed for operational safety reasons. Regarding initialization times, the real system requires 1 minute and 28 seconds for complete startup pipeline (compared to instantaneous activation of simulated software). This latency is due to technical times necessary for handshake among drivers, network configuration, and ROS2 middleware synchronization, parameters maintained conservatively to ensure connection stability.

Chapter 7

Conclusions and Future Developments

7.1 Summary of Results

The present thesis work concerned the design and implementation of an integrated robotic system for automated crop sensing. The primary objective was development of an architecture capable of executing dynamic movement planning for a six-axis manipulator, ensuring autonomous achievement of every kinematically admissible configuration. In parallel, the communication module between physical hardware and the stereoscopic sensor was realized, together with a path planning algorithm capable of performing precision scans regardless of spatial arrangement of plant organisms. Based on experimental tests conducted, the thesis objectives are considered fully achieved. The system demonstrated high reliability, whose main strengths can be summarized as follows:

- **Modular Architecture:** System decomposition into three independent macro-modules (communication, middleware integration, and scanning logic) ensures high maintainability and facilitates possible replacement of hardware or software components.

- **Adaptive Movement Planning:** Unlike static scanning systems or those based on predefined trajectories, the developed algorithm dynamically adapts the robot path as a function of three-dimensional perception of the surrounding environment.

- **High Spatial Coverage and Precision:** The synergy between stereoscopic vision and Dobot CR5's six-axis kinematics allows overcoming typical occlusion limitations of fixed sensors. Results show target coverage between 90% and 100% in nominal configurations.

- **Collision Avoidance:** Systematic integration of MoveIt2's Planning Scene ensures every trajectory is preventively validated against collisions, guaranteeing physical integrity of robot and crops.

It was further observed that scanning quality for plants in non-standard positions could be further enhanced by optimizing robot base positioning (e.g., via suspended mounting), thus extending useful work volume.

7.2 Future Development Perspectives

The realized system lays solid foundations for agricultural monitoring automation; however, several enhancement areas can be identified that could elevate platform effectiveness and autonomy.

7.2.1 Integration with Mobile Robotic Bases (AGV)

The primary limitation of the current prototype lies in the stationary nature of the manipulator, which restricts sampling to the arm's range of action. Natural project evolution provides for robot installation on a mobile base (AGV). Integration of `SLAM`[1] algorithms would enable the system to autonomously navigate between crop rows, extending monitoring to entire cultivated areas without manual intervention.

7.2.2 Path Planning Optimization and Active Vision

Implementation of *Active Vision* techniques would enable the robot to modify trajectory in real-time based on information content of acquired images. Integration of `NBV`[2] algorithms based on neural networks could drastically reduce acquisition number, autonomously identifying plant zones with greater morphological uncertainty.

[1] SLAM (Simultaneous Localization and Mapping) is a technique enabling a robot to build a map of an unknown environment and simultaneously localize itself within it.

[2] NBV (Next-Best-View) defines a class of algorithms that, given partial scanning, calculate the next sensor position maximizing new information acquisition, minimizing necessary shots.

7.2.3 Reconstruction Refinement via NeRF and Gaussian Splatting

From the data processing perspective, the system could benefit from leading-edge 3D reconstruction techniques such as `NeRF`[11][3] or `3D Gaussian Splatting`[7][4]. Such technologies would facilitate high-precision phenotypic analysis, such as fruit counting or leaf volume estimation, with fidelity superior to standard point clouds.

7.2.4 System Adaptability to Industrial Applications

Finally, the proposed architecture, thanks to abstraction provided by ROS2, proves readily exportable to industrial domains such as mechanical component quality control or predictive maintenance in unstructured environments. Algorithm independence from specific hardware renders this infrastructure a versatile framework for modern collaborative robotics.

[3] Neural Radiance Fields (NeRF) employ neural networks to represent complex 3D scenes as continuous density and color functions, enabling high-fidelity novel view synthesis.

[4] 3D Gaussian Splatting is a volumetric rendering technique representing 3D scenes through a set of gaussians, ensuring photorealistic reconstruction and real-time rendering times superior to traditional methods.

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